

PD Group:

Name:

ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2008

BIOLOGY
Higher 2

9747/02

Paper 2 Core Paper

~~12~~ September 2008 ~~Fri~~

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READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section AAnswer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

Section BAnswer any **one** question.

Write your answers on separate writing paper.

At the end of the examination,

1. fasten all your work securely together;
2. hand in your answers for Section A and Section B separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

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Section A

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Section B**Total**This document consists of ~~18~~ printed pages.

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Section A

Answer **all** the questions in this section.

QUESTION 1

Fig. 1.1 is an electron micrograph of a fibroblast. Fibroblasts are flattened, irregular-shaped connective tissue cells that secrete components of extracellular matrix such as hyaluronic acid and collagen.

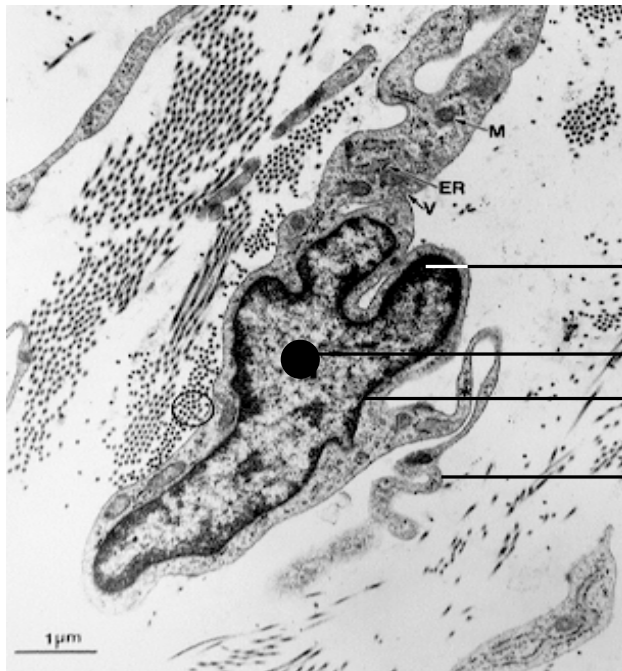


Fig. 1.1

Picture adapted from www.udel.edu/~histopage

(a) Name structures A to D.

A

B

C

D

(b) Explain why collagen is described as a fibrous protein.

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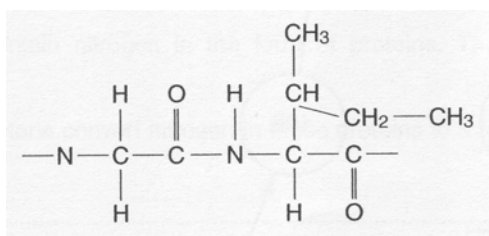
[2]

Collagen is continuously broken down in the extracellular matrix by the enzyme collagenase, which catalyses the hydrolysis of the peptide bond between the amino acids glycine and isoleucine.

- (c) Suggest how collagenase is only able to act on the peptide bond between glycine and isoleucine and not on peptide bonds between any other amino acids.

[2]

- (d) Draw a diagram below to show how the peptide bond between glycine and isoleucine is broken down by hydrolysis including the product or products.



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<#>Secondary structure (helical);¶
<#>Polypeptide chains cross link to form long fibres;¶
<#>Regular amino acid sequence / every 3rd residue is a glycine; ... [2]¶
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Deleted: Collagen is found in the extracellular matrix of muscles, tendons, ligaments and bones. Fibroblast cells in these tissues make collagen by synthesizing polypeptides that form molecules with a triple helix shape. These are secreted from fibroblasts into the extracellular matrix where enzymes assemble them into collagen fibres.¶

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¶ ... [19]

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(a) State two similarities betw ... [21]

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(c) Explain why collagen is d ... [22]

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QUESTION 2

Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a β -male.

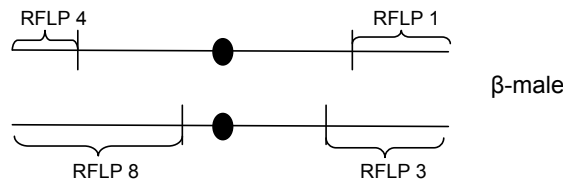


Fig. 2.1

An offspring of β -male has a chromosome with RFLPs 1 and 8, contributed by the father.

- (a) (i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8.

.....[1]

- (ii) In the space below, draw the pair of homologous chromosomes after the event that has occurred in (a)(i).

.....

- (iii) Suggest a reason why Huntington's Disease is likely to persist in populations.

.....

.....[1]

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Cell and Nuclear Division¶ ... [27]

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The micrograph below shows a lily cell undergoing cell division.



Fig.2.2 [© Clare Hasenkampf/BPS.]

(b) In reference to the micrograph above, identify:

(i) The type and stage of nuclear division

.....[2]

(ii) Briefly explain the reason for your answer in (b) (i).

.....[2]

(c) Describe the roles of spindle fibres in meiosis.

.....[2]

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QUESTION 3

An experiment was conducted to examine the effects of glucose and lactose on the levels of β -galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals. The results are shown in Fig. 3.1 below.

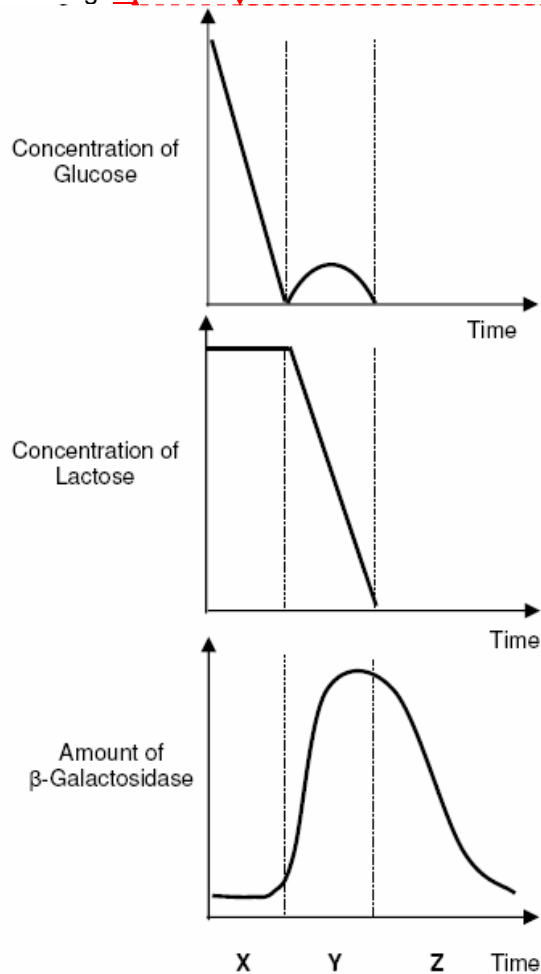


Fig.3.1

(a) State the two mechanisms of regulation of lac operon.

.....[2]

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TJC Prelim 2007

Deleted: Bacteria can regulate the expression of its genes in response to changes in the environment, typically to the presence or absence of nutrients. Some bacterial genes encode for regulatory proteins which controls the expression of other genes or operons. For example, *Escherichia coli* can utilize any sugar as its source of carbon and energy. However, when grown in a medium containing both glucose and lactose, the bacteria will use up one before the other. This preferential use of carbon source is said to be governed by the "dual control of the lac operon".

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<#>positive control by cAMP receptor protein; [2]
[1 mark each]

- (b) With reference to Fig. 3.1, explain how expression levels of β -galactosidase is regulated by the lac repressor from time X to Y.

X:

.....

.....

.....

.....

.....

.....[3]

Y:

.....

.....

.....

.....

.....

.....[3]

- (c) During period Z, suggest one mechanism that causes the amount of β -galactosidase to decrease.

.....[1]

- (d) In a separate experiment, *E.coli* were infected with T2 phage. Fig.3.2 shows the results of an experiment in which T2 was added to a culture of *E.coli*. Sample of the culture were then taken out at intervals to determine the number of free T2 virus present.

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 <#>In the presence of lactose, allolactose binds to lac repressor;¶
 <#>Thus, the Lac repressor is in its inactive configuration and does not bind to the operator;¶
 <#>But another mechanism prevents transcription of the lac operon due to catabolite repression (ie. CAP is inactive and disengages from CAP binding site);¶
 <#>RNA polymerase is unable to bind to the promoter/Low β -galactosidase mRNA is produced;¶
 [Any 3 - 1 mark each]¶

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 <#>Allolactose binds to the ... [65]

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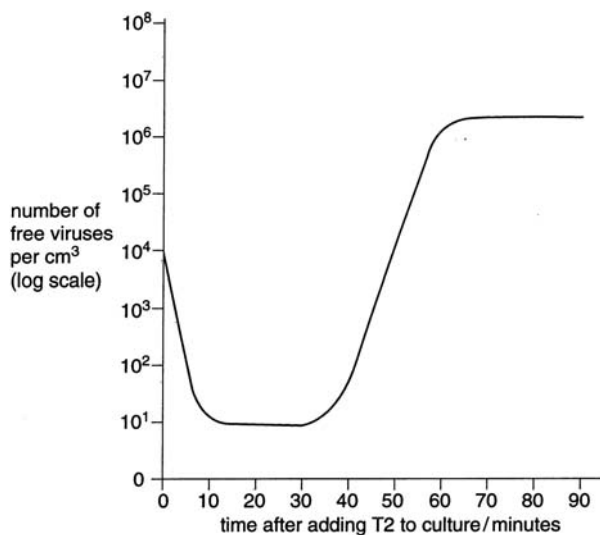


Fig. 3.2

With reference to Fig. 3.2, explain the changes in number of free T2 viruses

(i) In the first 10 mins;

.....

.....

.....[2]

(ii) between 30 and 60 minutes.

.....

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.....[3]

(e) Suggest with reasons, a possible evolutionary origin of viruses.

.....

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.....[2]

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Note : Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount¶

¶
(d) (i) Explain why luciferase activity can be used as a reflection of translation efficiency. [2]¶

.....¶
.....[2]¶

¶
<#>With reference to the Table 3.1, describe the effect of polyadenylation on mRNA degradation in capped mRNA. [2]¶

¶
.....¶
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[Total: 16]**QUESTION 4**

When a red flowered, three-leaved clover plant was crossed with a white flowered, five-leaved plant the following offspring were produced:

red-flowered, five leaved	36
red-flowered, three-leaved	41
white-flowered, five-leaved	40
white-flowered, three-leaved	38

The alleles for red flowers and five leaves are dominant.

(a)

- (i) Using the symbols **R** for red flowers, **r** for white flowers, **L** for five leaves and **l** for three leaves, draw a genetic diagram to explain these results. [5]

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¶

(d) (i) Luciferase (protein) is the end product of translation.¶

... easily detected as a light / measurable ... [2]¶

¶

(ii) Increase/ doubles mRNA half-life:¶

... from 53 units to 100 units; ... [2]¶

¶

(iii) Capping increases translation efficiency between 21 to 297 folds:¶

... Polyadenylation increases ... [84]

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(ii) When these plants were self-pollinated, explain why only the white-flowered, three-leaved plants bred true.

[2]

(b) Use the chi-squared (χ^2) test, and the table of probabilities shown in Table 1, to find the probability of the results of the cross in (a) departing significantly by chance from expectation. Show your working.

[2]

$$\chi^2 \text{ test} \quad \chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Key to symbols

Σ = 'sum of ...'

v = degrees of freedom

c = number of classes

O = observed 'value'

E = expected 'value'

Distribution of χ^2

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.60	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.34	16.27
4	7.78	9.49	11.67	13.28	18.46
5	9.24	11.07	13.39	15.09	20.52

Table 4.1

(i) Calculated $\chi^2 = \dots\dots\dots$ Probability = $\dots\dots\dots$ [1]

(ii) State the conclusion that can be drawn from your calculation of χ^2 .

$\dots\dots\dots$

$\dots\dots\dots$ [1]

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QUESTION 5

One way of investigating relationships between different primate groups is to study differences in the amino acid sequences of proteins such as haemoglobin.

(a) Explain the origin of differences in the amino acid sequences in a protein.

.....

.....

.....[2]

The β chain of haemoglobin has been investigated and shown to have only four points of difference in all primate groups. The amino acids found at these positions in certain primates or primate groups are shown in the table below.

Primate or primate group	Amino acid at position 80	Amino acid at position 87	Amino acid at position 109	Amino acid at position 125
Chimpanzees	Aspartic acid	Threonine	Arginine	Proline
Gibbons	Asparagines	Lysine	Arginine	Glutamine
Old World monkeys	Aspartic acid	Glutamine	Lysine	Glutamine
New World monkeys	Aspartic acid	Glutamine	Arginine	Glutamine
Lemurs	Aspartic acid	Glutamine	Threonine	Alanine

Table 5.1

(b) (i) With reference to Table 5.1, state the two groups that seem to be closely related and give a reason for your answer.

.....

.....[2]

(ii) Gibbons are usually classified with the chimpanzees as apes, in the Hominoidea. Explain why the results shown do not support this classification.

.....

.....

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.....[3]

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<#>changes the sequence of DNA bases / changes nucleotide / codon;¶
<#>new triplet codes for a different amino acid;¶
<#>substitution causes frameshift;¶
<#>codes for new polypeptide; [2]¶
[1/2 marks each]¶

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<#>Only one difference / three common amino acids;

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<#>Gibbon β chain has on[... [157]

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6 In a recent study, a family (Felidae) was found to have gone extinct. Within a few pockets of the population, it was suggested that this species in population most populated species at a very high.

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QUESTION 6

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Type 2 diabetes is a disease condition that is usually caused by the failure of target cells to respond to insulin. This is due to a defect in the insulin response mechanism of the target cells.

Fig. 6.1 shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.

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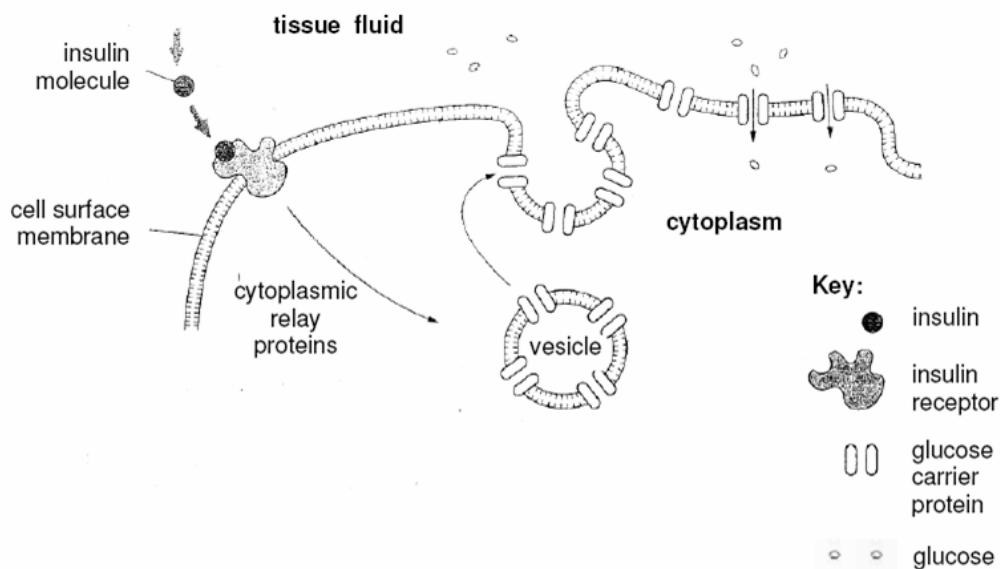


Fig. 6.1

(a) State the source of the insulin molecule in Fig. 6.1.

[1]

(b) The insulin receptor is a protein. Suggest how this may be a genetic risk factor for Type 2 diabetes.

[1]

insulin molecule
cell surface membrane

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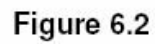
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(i) explain how the binding of insulin to its receptor results in activation of tyrosine kinase.

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QUESTION 7.

Fig. 7.1 is an electron micrograph which shows the axon of a motor neuron in transverse section.

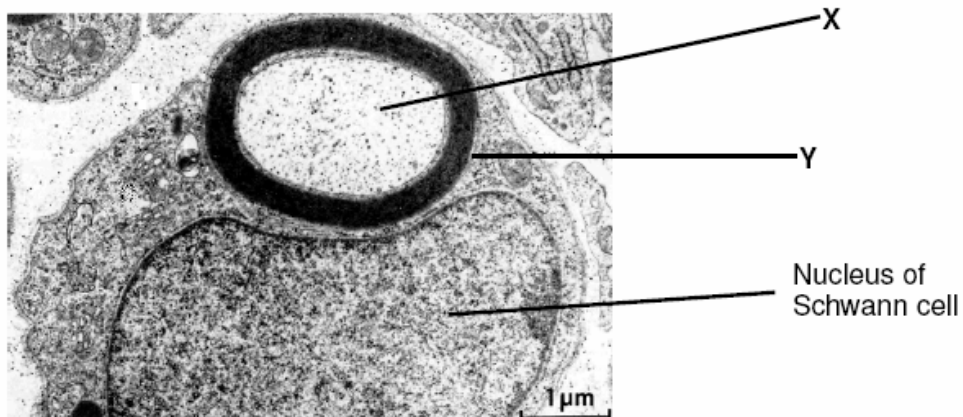


Fig. 7.1

(a) Label the structures X and Y.

X:

Y: [1]

(b) Explain the significance of Y in electrical transmission of nervous impulses.

.....

 [3]

Often present at the synaptic knob of many motor neurons is a large number of mitochondria.

(c)(i) State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor neurone.

.....

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.....[2]

(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.

.....[2]

(d) Predict the effect on an action potential when the external concentration of sodium ions is lowered.

.....[2]

.....[2]

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- (d) . Figure 2 below is a series of graphs showing intracellular and extracellular concentrations of sodium (Na⁺) and potassium (K⁺) ions in several cell types.<sp>¶
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Section B: Free Response Questions
Answer only **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

8. (a) Outline the role of NAD and NADP in cells. [7]
(b) Describe how photophosphorylation differs from oxidative phosphorylation. [7]
(c) Describe the roles of membranes in photosynthesis and in respiration. [6]
[Total: 20]

9. (a) Discuss why the structural features of the eukaryotic genome are considered to be more complex than those of the prokaryotic genome. [8]
(b) Compare and contrast the control of gene expression in prokaryotic and eukaryotic cells. [6]
(c) Describe the structure and function of the portions of eukaryotic DNA that do not encode for protein or RNA. [6]
[Total: 20]

Mr Ong, Mr Ang and Ms Resma wish you the BEST!

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QUESTION 9

a. Discuss why the

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2	Multiple chro - usually in dip - pairing of ho allows for gen meiotic form o allelic interac - gives rise to
3	Presence of i - presence of i - function as y gene regulati
4	Monocistroni - individual en by separate p - can be produ at all; - hence allowi

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Nuclear envelope/ membrane;

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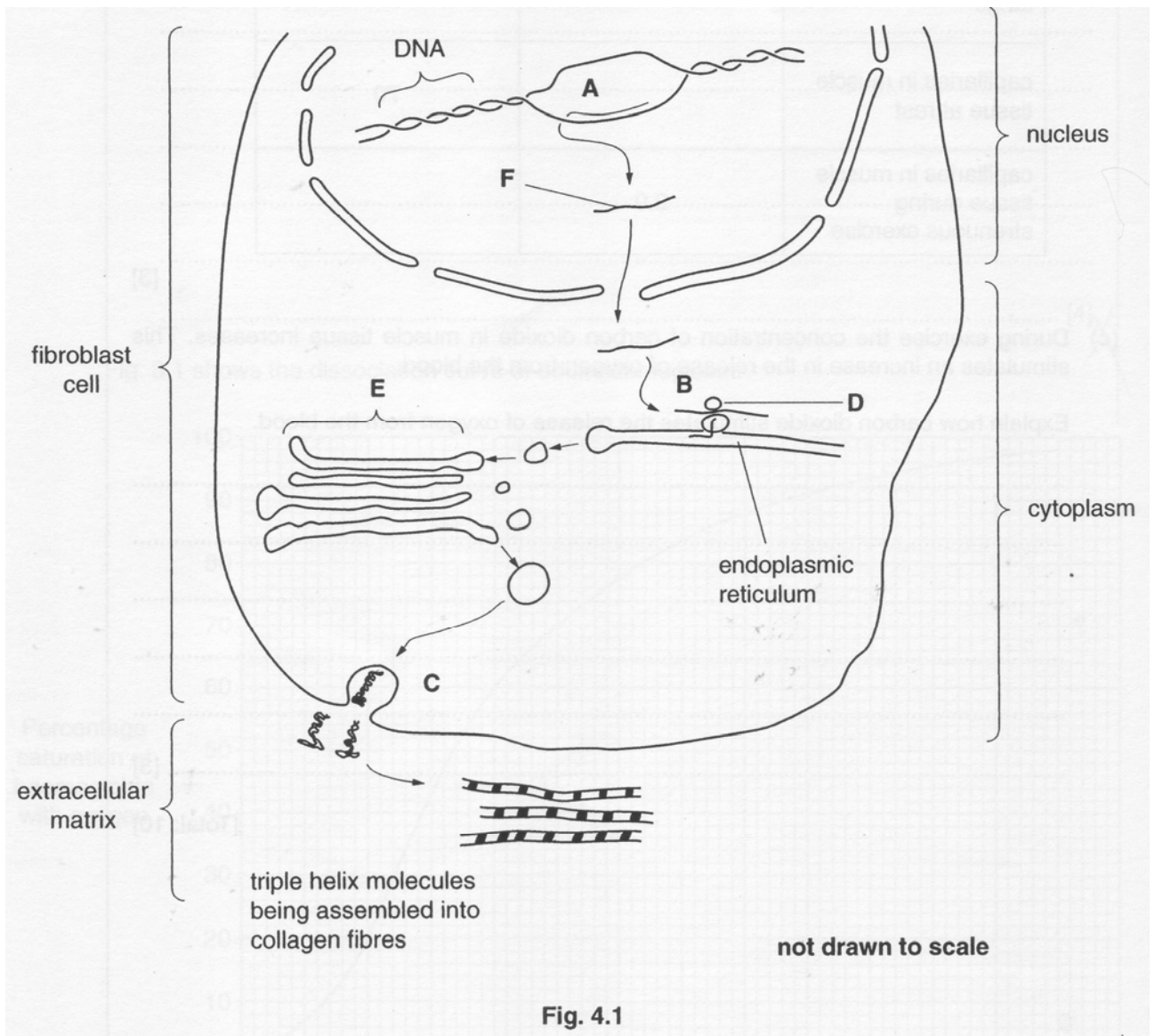
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Nov 2003 Paper 2 Q1

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Fig. 4.1 is a diagram summarizing these events.



(a) (i) Name the processes occurring at **A**, **B** and **C**. [3]

A Transcription
B Translation
C exocytosis

(ii) Name structures **D** and **E**. [2]

D Sub unit of ribosome
E Golgi apparatus/body

(iii) Name molecule **F**. [1]

F mRNA

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a- to remove diagram and combine with something else....

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VJC Promo 2006 Q1d

(a) State two similarities between rough endoplasmic reticulum and golgi apparatus.

[2]

(b) State two differences between rough endoplasmic reticulum and golgi apparatus.

[2]

Similarity (any one) - Both consist of membrane-bound sacs/tubules called cisternae. [1] - Both are made up of single membrane; [1] - Both produce vesicles that bud off the organelle; [1/2]	
Differences	
Structure D (rER)	Structure F (GA)
1. Consist of a <u>network</u> of <u>interconnected</u> membrane-bound sacs and tubules. [1/2]	Consist of <u>stacks</u> of <u>flattened</u> membrane-bound sacs. [1/2]
NB: idea of interconnectedness is imp.	
2. Presence of ribosomes. [1/2]	Absence of ribosomes. [1/2]
® ribosomes <i>may or may not</i> be present	
3. Connected to the outer membrane of the nucleus. [1/2]	3. Not connected to the outer membrane of the nucleus. [1/2]
4. Does not have a definite forming and maturing face.	4. Has a definite forming face where the vesicles join and a maturing face where the vesicles bud off.

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Nov 2003 Paper 2 Q1

(c) Explain why collagen is described as a fibrous protein.

[2]

insoluble nature of collagen;
 secondary structure – three polypeptides chains are cross-linked via hydrogen bonding to each other to form a triple helix;
 Fibrils like parallel in bundles to form a collagen fiber;

active site;
specific complementary shape;
only accepts R grps of these 2

amino acids; [2]
[1 mark each]

.....
.....

.....
.....

.....
.....

.....
.....

.....[3]

Correct bond broken (between C-N);
Involvement of water molecules in breaking the peptide bond shown
clearly;
Two amino acids with free grps as follows
-COOH/-COO⁻ and -NH₂/-NH₃⁺;
[3]
[1 mark each]

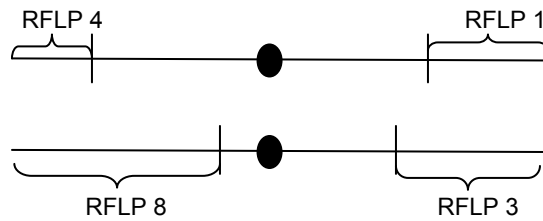
b and c are good :)

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Cell and Nuclear Division
RJC Prelim 2007

Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a male.



An offspring of the male has a chromosome with RFLPs 1 and 8, contributed by the father.

- (a)(i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8.
[1]

-
- (ii) In the space below, draw the event occurring in (a).
[3]

- (iii) Suggest a reason why Huntington's Disease is likely to persist in populations. [1]

The micrograph below shows a Lily cell undergoing cell division.



[© Clare Hasenkampf/BPS]

- (b) Refer to the micrograph above and:
- (i) Identify the type and stage of nuclear division. [2]

- (ii) Briefly explain the reason for your answer in (b)(i). [2]

(c) Describe the roles of spindle fibers in meiosis.
[2]

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crossing over;

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between non-sister chromatids

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at prophase I;
[1/2 marks each]
[1]

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Redraw and scan chromosome diagram.

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[1] - 2 sister chromatids per chromosome
[1] – a crossing over of non-sister-chromatids; may also take place on the other arm betw RFLP 4 and 8labelling of centromere (1/2) and chromatids/chromosomes (1/2)

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English (U.S.)

most individuals have children before they realize they have inherited the disease;

[1]

[1/2 marks each]

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[© Clare Hasenkampf/BPS.]

Meiosis;

Metaphase 2;

[1 mark each]

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metaphase because chromosomes are aligned at equator;

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2 metaphase plate has formed → meiosis;

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**the daughter cell have chromosomes aligned at the new equatorial plate
which is at right angles to the first;**

[Any 3 - 1 mark each]

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**pull sister chromatids/homologous chromosomes to opposite poles at
anaphase;**

help align chromosomes at equatorial plate;

[1 mark each]

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**Other (non-kinetochore) spindle fibres lengthen to elongate the cell in
preparation for cytokinesis**

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Once the glucose concentration has depleted, *E. coli* uses Lactose; Allolactose binds to the lac repressor; Lac repressor is inactive and does not bind to the operator; RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lacoperon / Lac operon is switched on / High β -galactosidase mRNA is produced;

[Any 3 - 1 mark each]

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X:

Y:

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Describe another mechanism which regulates the expression of β -galactosidase at Y.

[2]

(d)

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(i) S

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mRNA degradation/ protein is degraded.

[1]

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(ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without a carbon source.

[1]

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(a) State the two mechanisms of regulation of lac operon. [2]

negative control by the lac repressor
positive control by cAMP receptor protein.

(b) With reference to Fig.1, explain how expression levels of β -galactosidase is regulated by the **lac repressor** from time X to Y. [6]

X:

Both glucose and lactose are present and glucose is used but not lactose
In the presence of lactose, allolactose binds to lac repressor. Thus, the Lac repressor is in its inactive configuration and does not bind to the operator
But another mechanism prevents transcription of the lac operon due to catabolite repression (ie. CRP is inactive and disengages from CRP binding site)
RNA polymerase is unable to bind to the promoter/Low β -galactosidase mRNA is produced

Y:

Once the glucose concentration has depleted, *E. coli* uses Lactose. Allolactose binds to the lac repressor
Lac repressor is inactive and does not bind to the operator
RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lac operon / Lac operon is switched on / High β -galactosidase mRNA is produced

(c) Describe another mechanism which regulates the expression of β -galactosidase at Y. [2]

In low levels of glucose, cAMP levels increases , CRP is active and it binds to the CRP binding site.
Transcription of B-gal proceeds at a high level.

(d)(i) Suggest one mechanism that causes the amount of β -galactosidase to decrease. [1]

mRNA degradation/ protein is degraded

(ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without a carbon source. [1]

Stress genes can be turned on which allows the bacterial cells to survive.

Under

stress or starvation (starvation sigma factor)

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In an experiment to investigate the function(s) of capping and polyadenylation, scientists introduced firefly luciferase mRNA, with and without a cap, and with and without poly (A), into tobacco cells. The amount of remaining luciferase mRNA and the luciferase activity were measured experimentally. Luciferase is a protein product that is easy to detect because of the light it generates in the presence of appropriate substances (luciferin and ATP).

The table below illustrates the effects of capping and polyadenylation on the lifespan of the luciferase mRNA and its protein products.

Table 3.1

mRNA	Luciferase mRNA Half Life (min)	Luciferase Activity (units)
Uncapped		
Poly(A) tail absent	31	2,941
Poly(A) tail present	44	4,480
Capped		
Poly(A) tail absent	53	62,595
Poly(A) tail present	100	1,331,917

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Note : Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount

(d) (i) Explain why luciferase activity can be used as a reflection of translation efficiency. [2]

.....
.....
.....[2]

With reference to the Table 3.1, describe the effect of polyadenylation on mRNA degradation in capped mRNA. [2]

.....
.....
.....[2]

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With reference to Table 3.1, describe the effects of capping and polyadenylation on translation efficiency.

.....
.....
.....[2]

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**Capping increases translation efficiency between 21 to 297 folds;
Polyadenylation increases translation efficiency between 1.5 to 21 folds; [2]**

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- (d) (i) Luciferase (protein) is the end product of translation; easily detected as a light / measurable [2]
- (ii) Increase/ doubles mRNA half-life; from 53 units to 100 units; [2]
- (iii) Capping increases translation efficiency between 21 to 297 folds; Polyadenylation increases translation efficiency between 1.5 to 21 folds [2]

- (e) 5 Small nuclear Ribonuclear Proteins (snRNPs) associate with other proteins; form a **spliceosome**;
 RNA-protein interactions create a **lariat** structure;
Spliceosome recognizes the splicing signals/ splice site/ short nucleotide sequences at the end of intron on a pre-mRNA molecule;
 Lariat structure excises introns;
 Joins exons together to form mature mRNA;

c and d ii too specific but can use 1st part.

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Genetic Variation (Crossing)
 Nov 2006 Paper 2 Q4

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A cross was made between two fruit flies. One parent had an unknown genotype since it had the wild phenotype (normal [long] wings and normal body colour). The other parent had vestigial [undeveloped] wings and ebony body colour, so it was known to be homozygous recessive for both characteristics.

The resulting offspring were as follows:

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normal wings and normal body colour

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normal wings and ebony body colour

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vestigial wings and normal body colour

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vestigial wings and ebony body colour

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Using the following symbols:

N normal wings **n** vestigial wings
E normal body **e** ebony body

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Draw a genetic diagram in the space below to show the cross described.

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Parental phenotype	Red flowered, three-leaved clover plant normal [long] wings and normal body colour	White flowered, five-leaved plant vestigial [undeveloped] wings and ebony body colour
Genotype	Unknown (NnEe)Rrll	rrLlnnee;
meiosis		
Gametes		
meiosis		
Gametes		
	NnEeRrLI Red flowered, five -leaved	Nnee rrLinnEe White flowered, five -leaved Nnee;
	Rrll Red flowered, three -leaved	rrll; White flowered, three -leaved

F1 genotype	1 RrLl1 NnEe	1 Rrll	1 Rrll1 Nnee 1 nnEe	1 rrLl ;1 nnee;
F1 phenotype	1 Red flowered, five -leaved normal wings and normal body colour	1 Red flowered, three -leaved	1 White flowered, five - leaved normal wings and ebony body colour vestigial wings and normal body colour	1 White flowered, three - leaved ;vestigial wings and ebony body colour;

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Explain why it would be useful to carry out a chi-squared test on these results. No calculations are required to answer this question.

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Expected results is a ratio of 1:1:1:1;

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Whether observed results differed significantly from expected results with a less than 5%;

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**white-flowered, three-leaved plants were double homozygous recessive;
All the other plants would have been heterozygotes;**

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Vestigial wing and ebony body are caused by artificially induced mutations in laboratory bred fruit flies. Suggest why such mutants are unlikely to be found in natural populations.

[2]

Vestigial wing and ebony bodied flies would be disadvantaged in the wild because they cannot fly from and are more visible to predators and therefore unlikely to survive as they are more likely to be eaten;
Mutations are rare and such alleles are not passed on to offspring because they are unlikely to reproduced;
Such characters are coded for by recessive alleles which are therefore masked by dominant alleles;

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Penotype	Observed (O)	Expected (E)	$(O-E)^2$	$(O-E)^2/E$
Red flowered, five -leaved	36	38.75	7.5625	0.195
Red flowered, three -leaved	41	38.75	5.0625	0.131
White flowered, five -leaved	40	38.75	1.5625	0.040
White flowered, three -leaved	38	38.75	0.5625	0.015
				$\chi^2 = 0.381;$

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degrees of freedom = $n - 1$

$$= 4 - 1 = 3;$$

Calculated $\chi^2 = 0.381$
Critical $\chi^2 = 7.82$;

Probability that observed results deviated from expected due to chance is high (0.05 level of significance);

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Therefore accept null hypothesis. Observed results are not significantly different from expected results;

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**Gibbon more like New World monkey;
Gibbon β chain has only one common amino acid (arginine) with chimpanzee;
2 common amino acids with New World monkeys;
Chimpanzee has only common amino acid with / 3 different from each of the other groups;**

[3]

[Any 3 - 1 mark each]

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two

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**ref to homology / skeletal / skull / dentition details;
dating of fossils elaborate?;
anatomical comparisons of fossil primates;
anatomical comparisons of the skeletons of modern species;
sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting
antibody-antigen / immunological / serum studies;
use of antibodies / agglutination/ immuno-precipitation; isn't this same as previous pt?**

[Any 4 - 1 mark each]

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Answers

- (a) caused by point mutation;
changes the sequence of DNA bases / changes nucleotide / codon;
new triplet codes for a different amino acid;
substitution causes frameshift;
codes for new polypeptide [2]

(b)(i) Old World monkeys and New World monkeys;
Only one difference / three common amino acids [2]

(b)(ii) Gibbon more like New World monkey;
Gibbon β chain has only one common arginine / amino acid with chimpanzee;
2 common amino acids with New World monkeys;
chimpanzee has only common amino acid with / 3 different from each of the other groups [3]

(c) anatomical comparisons of fossil primates;
dating of fossils;
sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting
anatomical comparisons of the skeletons of modern species;
ref to homology / skeletal / skull / dentition details;
antibody-antigen / immunological / serum studies;
use of antibodies / agglutination/ immuno-precipitation; [any 2]

modify the peak for fig. 6.2 at 1st part...

6 What do you understand by the following terms?

- (a) i. *Species* [2]
- Any group of organisms;
 - Capable or potentially capable of interbreeding with one another;
 - To produce fertile offspring;
 - Show any similarities in evolution, ecology, morphology, behavior, biochemistry (any 2);
- ii. *Genetic variation* [1]
- Members of a population;
 - Differ in one or more inherited characteristics;
- (b) What impact would low genetic variability in a population of this species have on species survival? Explain your answer. [2]
- Low number of alleles would reduce probability of the presence of a favorable characteristic in the population;
 - If there was an environmental change;
 - Reduced chances of survival / species extinction;
 - Continuous inbreeding between closely related individuals;
 - Inbreeding depression may occur;
- (c) Using the information from **Figure 6.1**, explain the variation shown. [2]
- Polygenes;
 - Continuous variation / normal distribution / bell shaped;
 - Range seen due to the additive effects of genes / alleles;
 - Effects of environment;
- (d) How does the information in **Figure 6.2** support your answer in (c)? [1]
- Breeding with individuals of similar genotypes;
 - Shifted the mean / variation towards either end;

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Answer:

(a)

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Identify the source of the insulin molecule in **Figure 6.1**. [1]

Beta

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cells of the pancreatic islets of Langerhans;;
(**Reject:** b-pancreatic cells / b-islets)

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(b) The insulin receptor is a protein.

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be a genetic risk factor for Type 2 diabetes.

[1]

Harmful mutation of the **gene encoding the insulin receptor** (reject: insulin) may occur;

(Accept: reference to harmful effects of the mutation eg. alteration of specific 3-D conformation of insulin receptor or defective / non-functional / malfunctional receptor;

Reject: genetic defect / defective gene / faulty gene ie word "mutation" not mentioned.) mutant allele / gene is **transmitted to / inherited by future generations**;

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With reference to **Figures 6.1** and **6.2**,

(i)

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explain how the binding of insulin to its receptor results in activation of tyrosine kinase

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in an increase in the phosphorylation of serine and threonine residue side-chains of cytoplasmic relay proteins

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Binding of insulin causes the two receptor subunits / polypeptide chains / monomers / molecules (reject: two receptors) to form a dimer / dimerise ; dimerisation activates the tyrosine kinase region of each subunit;		
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leading to (auto)phosphorylation of tyrosine residues on the tail of subunits;		
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Specific **relay protein(s)** in the cell **bind(s)** to specific **phosphorylated tyrosine(s)**; and becomes **activated**;

activated relay protein(s) trigger(s) **transduction pathway(s) / phosphorylation cascade**;

leading to **activation of** serine and threonine **kinases**; / **inhibition of** serine and threonine **protein phosphatases**;

in the cytoplasm. (Accept: serine and threonine kinases / protein phosphatases bind to phosphorylated tyrosine residues; and are activated; / inhibited; respectively) (max 1.5m)

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of serine and threonine residues of cytoplasmic relay proteins		
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that lead to an increase in the number of glucose channels in the cell surface membrane. [2]		
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Phosphorylation of cytoplasmic relay proteins constitutes a transduced chemical signal (accept: mention of p)		
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Phosphorylation cascade / downstream proteins activated / signal transduction / final activated relay proteins);		
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that is amplified; causing vesicles carrying glucose carrier proteins / transporters (reject: glucose channels); to move / migrate (reject: diffuse) towards cell surface membrane; and fuse (accept: mention of exocytosis; reject: bind) with it; carrier proteins are then incorporated / inserted into or embedded in membrane (reject: released / added to membrane) as glucose channels;		
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causing vesicles carrying **glucose carrier proteins / transporters** (reject: glucose channels);

to **move / migrate** (reject: diffuse) towards cell surface membrane;

and **fuse** (accept: mention of exocytosis; reject: bind) with it; carrier proteins are then **incorporated / inserted into or embedded in membrane** (reject: released / added to membrane) as glucose channels;

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Type 2 diabetes patients are not able to regulate their blood glucose levels owing to the loss of their cells' sensitivity to insulin.		

(i)

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If an increase in blood glucose concentration is not corrected for some time, symptoms of diabetes such as excessive thirst may occur. Suggest how excess blood glucose may cause excessive thirst. [1]		

Glucose causes **water potential** of the **blood to become more negative** (reject: lower / decrease); **OR**
so that water moves out of cells

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/ tissue fluid into the blood / loss of water from cells by (ex)osmosis ;		

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leading to excessive thirst.		

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(Accept: excessive urination; / increase in blood osmolarity; stimulates hypothalamus or brain;)		

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Explain how the blood insulin level is homeostatically regulated in a normal individual.

[3]

Hyperglycaemia / increased blood glucose concentration stimulates the **secretion / release of insulin**, the level of which rises;

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(note: both primary stimulus and response must be mentioned before 0.5m awarded)

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Insulin **accelerates / promotes / enhances glucose uptake / absorption**

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in target cells;

The consequent **decrease in blood glucose concentration** back to normal levels;

exerts **negative feedback**; (note: negative feedback must be mentioned in the context of controlling blood insulin, not glucose, level before 0.5m awarded) on the pancreatic _ cells to **inhibit insulin secretion / release**;

cells activated to secrete **glucagon** which **inhibits insulin secretion / release**; blood insulin level then falls.

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concentration of sodium ions are

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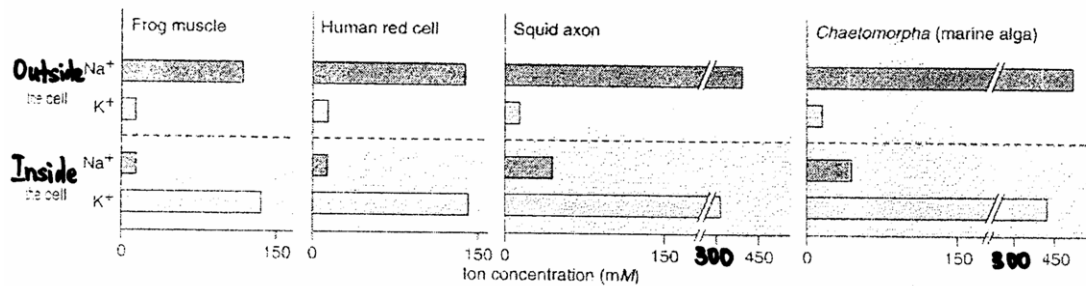
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(d) Figure 2 below is a series of graphs showing intracellular and extracellular concentrations of sodium (Na^+) and potassium (K^+) ions in several cell

types.



The double stroke (//) in the graphs of squid axon and *Chaetomorpha* (marine alga) shows that the graphs are cut.

Figure 2

With respect to Figure 2, state the trends observed in all cell types.
[2]

- (ii) Which part of the cell allows for the generation of these trends stated in (a) (i)

[1]

- (iii) Suggest a reason why the maintenance of this distribution of Na⁺ and K⁺ ions is important to these cell types.
[1]

- (c) Describe the events that lead to the release of the neurotransmitter from the presynaptic membrane into the synaptic cleft.

[2]

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Answers:

Figure 2 is an electron micrograph which shows the axon of a motor neurone in transverse section.

(a)

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Label the structures X and Y. [2]		

X: axoplasm Y: myelin sheath

(b)

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Explain the significance of Y in electrical transmission of nervous impulses. [3]		
Y acts as an electrical insulator, preventing the loss of current from the axon during electrical transmission.		
In a myelinated neurone, action potentials are regenerated only at the nodes of Ranvier / propagated by saltatory conduction, thus increasing the speed of electrical transmission.		
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Often present at the synaptic knob of many motor neurons		
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are a large number of mitochondria.		

(c)(i)

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of the neurone.		
ATP is used for the transport / movement of synaptic vesicles towards the membrane of the synaptic knob for the release of neurotransmitter.		
ATP used for the absorption / endocytosis of choline and acetate back into the		
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State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor

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neurone. [2]

Any two:

ATP is required for the Na⁺/K⁺ pump to work. Na⁺ ions are pumped out while K⁺

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ions are taken into the neurone via active transport.

ATP is required for the action of Ca²⁺ pumps which actively transport Ca²⁺ ions out

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(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.

[2]

During intense activity, more action potentials are generated;

Increased activities of Na⁺ - K⁺ pumps needing more ATP

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Superscript

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(d) Suggest the effect on an action potential when the external concentration of sodium ions are lowered.

[2]

Less sodium ions enter the neurone;
Hence lower action potential

(d) (i)

High Na^+ concentration outside the cell and a high K^+ concentration inside the cell;
Quote any cell's values –

Location	Ions	Ions concentration in			
		Frog muscle	Human r.b.c	Squid axon	<i>Chaetomorpha</i> (marine alga)
Outside the cell	Na^+	114	150	400	500
	K^+	14	14	14	14
Inside the cell	Na^+	14	14	42	42
	K^+	135	150	300	400

[Each point → 1 mark]

(ii)

Cell membrane;
 Na^+ and K^+ ion channels/ Na^+/K^+ pump;

[Each point → ½ marks]

(iii)

Maintains the resting membrane potential to generate nerve impulse;

OR

Contributes to the electrochemical gradient to establish the resting membrane potential;

[Each point → 1 mark]

(c)

Action potential arrives at the synaptic knob;
Depolarization increases membrane permeability to Ca^{2+} ;
opening of voltage gated calcium channels and Ca^{2+} influx;
 Ca^{2+} influx causes synaptic vesicles to fuse with presynaptic membrane and discharge neurotransmitters by exocytosis;

Underlined words should appear in student's answer.

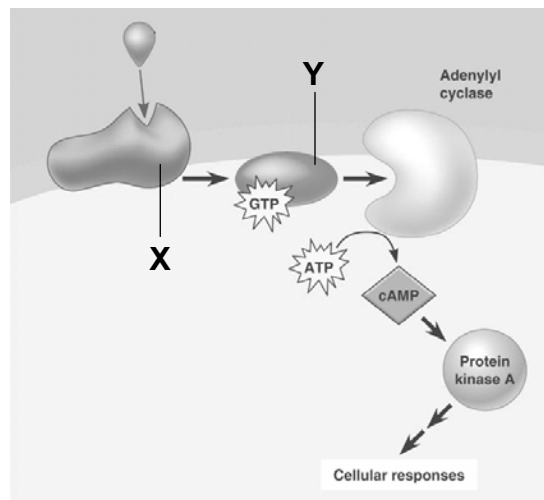
[Each point → ½ marks]

Cell Signaling
SAJC Prelim 2007
QUESTION 8

Certain microbes cause disease by disrupting G-protein signaling pathways.

The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human feces colonizes the small intestine and produces a toxin that modifies a G protein.

The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salts into the intestines, leading to profuse diarrhea and death from loss of water and salts.



(a) Name molecules X and Y.

X Y[1]

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways?

.....[1]

(c) What is the significance of a phosphorylation cascade in cell signaling?

.....
.....
.....

.....

.....[2]

(d) State the function of adenylyl cyclase.

.....[1]

(e) Suggest how the toxin leads to an increase in cAMP concentration.

.....

.....

.....

.....[3]

[Total: 8]

(a) Name molecules X and Y. [1]

[½ mark per pt]

X = G-protein linked receptor

Y = G-protein

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways? [1]

second messengers

(c) What is the significance of a phosphorylation cascade in cell signaling? [2]

signal amplification

specificity of cell signaling

ref. cross-talk (interaction between different signaling pathways)

(d) State the function of adenylyl cyclase. [1]

convert ATP to cAMP

(e) Suggest how the toxin leads to an increase in cAMP concentration.

[3]

modified G protein unable to hydrolyze GTP to GDP

remains stuck in its active form

adenylyl cyclase is continuously stimulated to make cAMP

Photosynthesis and Respiration
Nov 2006 Paper 2 Q7

NADP/ NAD as

- Electron/hydrogen carrier or coenzymes;
 - Carry hydrogen atoms from one compound to another;
- [max1, ½ m each]

NADP in photosynthesis

- Final electron acceptor in non-cyclic photophosphorylation;
 - Drives photolysis of water;
 - $2\text{H}^+ + 2\text{e}^- + \text{NADP} \rightarrow \text{NADPH}$ or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
 - NADPH passes to Calvin cycle/ light independent reactions;
 - in the stroma;
 - NADPH reduces glycerate biphosphate to triose phosphate and gets reoxidized back to NADP;
- [max2, ½ m each]

NAD in respiration

Aerobic respiration:

Glycolysis: Accept H^+ from triose phosphate, which is oxidized to pyruvate;
Forms 2 NADH for each glucose molecule;
Link reaction: Accept H^+ from pyruvate, which is converted to acetyl CoA;
Forms 2 NADH for each glucose molecule;
Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;
Forms 6 NADH for each glucose molecule;
Link rxn and Krebs cycle occurs in the mitochondrial matrix
Carry hydrogen/ electrons to ETC;
Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;
1 NADH yields 3 ATP;
NAD reoxidised;

Anaerobic respiration:

NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;

**NAD reused in glycolysis to generate 2 ATP;
[max5, ½ m each]**

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**Coenzyme NAD and NADP are reduced, carry electrons and protons to the electron transfer chain and are regenerated;
NAD is formed in glycolysis, link reaction and Krebs's cycle by oxidation;
in the mitochondrial matrix;
And then used in oxidative phosphorylation;
NADP location is in the stroma of the chloroplast;
NADP is reduced in the light reaction as the final electron acceptor;
And then was used in the Calvin cycle as a reducing agent;**

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Feature	photophosphorylation	oxidative phosphorylation
Organelle	Chloroplast	Mitochondria
Location	Thylakoid membrane	Inner Mitochondrial membrane
Proton reservoirs	Thylakoid space	Intermembrane space
Energy conversion	Light to chemical	Chemical to chemical
Role of coenzymes	NADP is reduced in the light rxn as final electron acceptor	NAD is reduced in glycolysis, link rxn and Kreb's cycle
Sources of electrons	Water	NADH and FADH
Roles of O ₂	By-product of photolysis	Final electron acceptor
Roles of water	Photolysis: To produce H ⁺ which will then accumulate in the thylakoid space and electrons to replace the electron lost at the primary reaction centre	By-product of O ₂ as the final electron acceptor

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Nov 2002

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Photosynthesis

Photosystems / antennae complex / chlorophyll in thylakoid membranes;
ETC, stalked particles (ATP synthase) in thylakoid membranes;
Thylakoid membranes are ion-impermeable / ref. to H^+ gradient;
Ref. to thylakoid membrane separating compartments (compartmentalization);
Ref. to double membrane / envelope;
Ref. to enzymes and Calvin cycle in the stroma; [max 3]

Respiration

Folding of cristae to increase surface area so that there are more stalked particles;
More ETC for oxidative phosphorylation;
ETC and stalked particles in inner membrane / cristae;
Inner membrane ion impermeable / ref. to proton gradient;
Ref. to intermembrane space acting as reservoir of H^+ ;
Ref. to outer membrane as a barrier;
Ref. to enzymes and Krebs cycle in matrix [max 3]

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Explain the effect of carbon dioxide and light as limiting factors on the rate of photosynthesis.

CO₂ is a raw material for the calvin cycle

Its increased concentration increased the rate of photosynthesis

Its level is low (0.04%) in the atmosphere and therefore plays a major role in limiting photosynthesis

Light intensity is also important in the light dependent stage to excite electrons

Chl a absorbs max light at 680nm while chl b absorbs max light at 480nm

Compensation point is the light intensity when rate of photosynthesis and respiration is the same and it is reached at low light intensity

Is therefore rarely limiting during daylight hours

High light intensity also opens the stomata

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Nov 2002 Essay 10c roles of mebsms in PS and RS (6)

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Organization and Control of Pro- and Eukaryotic Genome		

HCI Prelim 2007

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Explain how homology (anatomical, embryological, molecular) supports Darwin's theory of natural selection.

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Discuss why the structural features of the eukaryotic genome are considered to be more complex than those of the prokaryotic genome.

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Describe the neutral theory of molecular evolution.

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(b) (c) Compare and contrast the control of gene expression in prokaryotic and eukaryotic cells.

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Explain how the substitution of one nucleotide in DNA can be selected for or against in an area endemic with malaria.

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Explain how changes at the genome level can lead to cancer.

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10 (b) Explain how a point mutation in the gene coding for a growth factor receptor can lead to cancer. [6]

Mutation in the proto-oncogene that codes the growth factor receptor, results in it becoming an oncogene/ becoming oncogenic

A point mutation results in a change in triplet code/ which changes the amino acid sequence in the growth factor receptor.

The mutated growth factor receptor will have different conformation and function differently.

The mutated receptor might be constitutively activated regardless of the presence of its ligand

OR the mutated receptor might be constitutively active because the mutated receptor does not dissociate from its ligand readily.

This results in persistent intracellular signaling for cell growth and proliferation leading to uncontrolled cell division and formation of tumors the increase in cell division increased chances of multiple mutations in other critical genes like tumor suppressor genes, DNA repair genes, cell cycle control genes

This results in cells becoming cancerous

Cancerous cells metastasize /spread to other parts of the body to form secondary tumors.

(Cancerous cells can deprive surrounding cells/organs of nutrients and oxygen and cause functional failure of invaded tissues.)

OR

Explain the role of telomeres in maintaining the integrity of a eukaryotic organism. [6]

Are non-coding DNA sequences found at the ends of chromosomes.

Prevents fusion between exposed ends of chromosomes - chromosomal mutation.

Which can disrupt regulatory control of genes on the adjoined chromosomes Prevents the activation of DNA repair enzymes which will recognize any exposed chromosomal ends as a defect and degrade the chromosomes.

serves as buffer DNA to protect the organism's genes from being eroded after successive rounds of DNA replication

so that critical proteins will still be synthesized in the daughter cells despite the shortened chromosomes

Cell will also undergo apoptosis after a limited number of cell division/mitosis (~50)/when a critical length of the telomere is reached

This limits the extent of accumulated mutations and prevents the development of cancer.

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